

## Hands On 2: Difference between JPA, Hibernate and Spring Data JPA:

MySQL:

USE ormlern;

```
CREATE TABLE employee (  
    em_id INT AUTO_INCREMENT PRIMARY KEY,  
    em_name VARCHAR(100) NOT NULL  
);
```

Employee.java:

```
package com.cognizant.ormlearn.model;
```

```
import jakarta.persistence.Column;  
import jakarta.persistence.Entity;  
import jakarta.persistence.GeneratedValue;  
import jakarta.persistence.GenerationType;  
import jakarta.persistence.Id;  
import jakarta.persistence.Table;
```

```
@Entity
```

```
@Table(name="employee")
```

```
public class Employee {
```

```
    @Id
```

```
@GeneratedValue(strategy = GenerationType.IDENTITY) // Handles MySQL  
AUTO_INCREMENT
```

```
@Column(name="em_id")
```

```
private Integer id;
```

```
@Column(name="em_name")
```

```
private String name;
```

```
// Default Constructor
```

```
public Employee() {}
```

```
// Parameterized Constructor for easy testing
```

```
public Employee(String name) {
```

```
    this.name = name;
```

```
}
```

```
// Getters and Setters
```

```
public Integer getId() { return id; }
```

```
public void setId(Integer id) { this.id = id; }
```

```
public String getName() { return name; }
```

```
public void setName(String name) { this.name = name; }
```

```
@Override
```

```
public String toString() {
```

```
    return "Employee [id=" + id + ", name=" + name + "];"
```

```
}  
}
```

### EmployeeRepository.java:

```
package com.cognizant.ormlearn.repository;  
  
import org.springframework.data.jpa.repository.JpaRepository;  
import org.springframework.stereotype.Repository;  
import com.cognizant.ormlearn.model.Employee;  
  
@Repository  
public interface EmployeeRepository extends JpaRepository<Employee, Integer> {  
}
```

### EmployeeService.java:

```
package com.cognizant.ormlearn.service;  
  
import org.springframework.beans.factory.annotation.Autowired;  
import org.springframework.stereotype.Service;  
import org.springframework.transaction.annotation.Transactional;  
import com.cognizant.ormlearn.model.Employee;  
import com.cognizant.ormlearn.repository.EmployeeRepository;  
  
@Service  
public class EmployeeService {  
  
    @Autowired  
    private EmployeeRepository employeeRepository;
```

```
@Transactional

public void addEmployee(Employee employee) {

    employeeRepository.save(employee);

}

}
```

### OrmLearnApplication.java:

```
package com.cognizant.ormlearn;

import java.util.List;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.ApplicationContext;
import com.cognizant.ormlearn.model.Country;
import com.cognizant.ormlearn.model.Employee;
import com.cognizant.ormlearn.service.CountryService;
import com.cognizant.ormlearn.service.EmployeeService;

@SpringBootApplication

public class OrmLearnApplication {

    private static final Logger LOGGER =
        LoggerFactory.getLogger(OrmLearnApplication.class);
```

```
private static CountryService countryService;

private static EmployeeService employeeService;


public static void main(String[] args) {

    ApplicationContext context = SpringApplication.run(OrmLearnApplication.class, args);

    LOGGER.info("Inside main");


    // Fetch Beans from Context

    countryService = context.getBean(CountryService.class);
    employeeService = context.getBean(EmployeeService.class);


    // Run Tests

    testGetAllCountries();
    testAddEmployee();
}


private static void testGetAllCountries() {

    LOGGER.info("Start - Country Test");

    List<Country> countries = countryService.getAllCountries();

    LOGGER.debug("countries={}", countries);

    LOGGER.info("End - Country Test");
}


private static void testAddEmployee() {

    LOGGER.info("Start - Employee Add Test");
```

```
// Creating a test dummy employee instance

Employee newEmp = new Employee("Aditya Yadav");


LOGGER.debug("Saving employee: {}", newEmp.getName());
employeeService.addEmployee(newEmp);


// The ID gets populated automatically after save due to GenerationType.IDENTITY
LOGGER.info("Employee saved successfully! Generated ID: {}", newEmp.getId());
LOGGER.info("End - Employee Add Test");
}
}
```

## Output:

```
PS D:\SD\Digital-Nurture-JavaFSE-main\local_repo\Cognizant-exercercises\Week 2\Spring Data JPA\Exercise-2> & 'C:\Program Files\Eclipse Adoptium\jdk-25.0.3.9-hotspot\bin\java.exe' '@C:\Users\adity\AppData\Local\Temp\cp_xz3f5nf59on7cbs2viunqk3x_argfile' 'com.cognizant.orm_learn.OrmlLearnApplication'

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      | | | |
      |_| |_|

:: Spring Boot ::                (v4.1.0)

29-06-26 23:24:09.999 restartedMain      INFO c.c.o.OrmlLearnApplication      logStarting      54 Starting OrmlLearnApplication using Java 25.0.3 with
PID 8820 (D:\SD\Digital-Nurture-JavaFSE-main\local_repo\Cognizant-exercercises\Week 2\Spring Data JPA\Exercise-2\orm-learn\target\classes started by adity in
D:\SD\Digital-Nurture-JavaFSE-main\local_repo\Cognizant-exercercises\Week 2\Spring Data JPA\Exercise-2)
29-06-26 23:24:10.001 restartedMain      DEBUG c.c.o.OrmlLearnApplication      logStarting      55 Running with Spring Boot v4.1.0, Spring v7.0.8
29-06-26 23:24:10.002 restartedMain      INFO c.c.o.OrmlLearnApplication      logStartupProfileInfo 652 No active profile set, falling back to 1 default pr
ofile: "default"
29-06-26 23:24:10.037 restartedMain      INFO ertyDefaultsPostProcessor      log      69 Devtools property defaults active! Set 'spring.dev
tools.add-properties' to 'false' to disable
29-06-26 23:24:10.331 restartedMain      INFO toryConfigurationDelegate      registerRepositoriesIn 145 Bootstrapping Spring Data JPA repositories in DEFAU
LT mode.
29-06-26 23:24:10.371 restartedMain      INFO toryConfigurationDelegate      registerRepositoriesIn 213 Finished Spring Data repository scanning in 33 ms.
Found 2 JPA repository interfaces.
29-06-26 23:24:10.575 restartedMain      INFO org.hibernate.orm.jpa      logPersistenceUnitInformation 102 HH#008540: Processing PersistenceUnitInfo [name
: default]
29-06-26 23:24:10.612 restartedMain      INFO org.hibernate.orm.core      logVersion      41 HH#000001: Hibernate ORM core version 7.4.1.Final
29-06-26 23:24:10.868 restartedMain      INFO SpringPersistenceUnitInfo      addTransformer      129 No LoadTimeWeaver setup: ignoring JPA class transfo
rmer
29-06-26 23:24:10.887 restartedMain      INFO c.z.h.HikariDataSource      getConnection      109 HikariPool-1 - Starting...
29-06-26 23:24:11.116 restartedMain      INFO c.z.h.pool.HikariPool      checkFailFast      580 HikariPool-1 - Added connection com.mysql.cj.jdbc.C
onnectionImpl@21ac8a6d
29-06-26 23:24:11.118 restartedMain      INFO c.z.h.HikariDataSource      getConnection      122 HikariPool-1 - Start completed.
29-06-26 23:24:11.176 restartedMain      INFO o.h.o.connections.pooling      logConnectionInfo 179 HH#10001005: Database info:
Database JDBC URL [jdbc:mysql://localhost:3306/ormlearn]
Database driver: MySQL Connector/J
Database dialect: MySQLDialect

PS D:\SD\Digital-Nurture-JavaFSE-main\local_repo\Cognizant-exercercises\Week 2\Spring Data JPA\Exercise-2> & 'C:\Program Files\Eclipse Adoptium\jdk-25.0.3.9-hotspot\bin\java.exe' '@C:\Users\adity\AppData\Local\Temp\cp_xz3f5nf59on7cbs2viunqk3x_argfile' 'com.cognizant.orm_learn.OrmlLearnApplication'

Database JDBC URL [jdbc:mysql://localhost:3306/ormlearn]
Database driver: MySQL Connector/J
Database dialect: MySQLDialect
Database version: 8.0.46
Default catalog/schema: ormlearn/undefined
Autocommit mode: undefined/unknown
Isolation level: REPEATABLE_READ [default REPEATABLE_READ]
JDBC fetch size: none
Pool: DataSourceConnectionProvider
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown
29-06-26 23:24:11.662 restartedMain      INFO org.hibernate.orm.core      initiateService      53 HH#000489: No JTA platform available (set 'hibernat
e.transaction.jta.platform' to enable JTA platform integration)
29-06-26 23:24:11.696 restartedMain      INFO rEntityManagerFactoryBean      buildNativeEntityManagerFactory 464 Initialized JPA EntityManagerFactory for pers
istence unit 'default'
29-06-26 23:24:11.762 restartedMain      INFO .q.QueryEnhancerFactories      <clinit>      48 Hibernate is in classpath; If applicable, HQL parse
r will be used.
29-06-26 23:24:11.938 restartedMain      INFO c.c.o.OrmlLearnApplication      logStarted      60 Started OrmlLearnApplication in 2.2 seconds (process
running for 2.472)
29-06-26 23:24:11.940 restartedMain      INFO c.c.o.OrmlLearnApplication      main      24 Inside main
29-06-26 23:24:11.941 restartedMain      INFO c.c.o.OrmlLearnApplication      testGetAllCountries      36 Start - Country Test
29-06-26 23:24:12.067 restartedMain      DEBUG org.hibernate.SQL      logStatement      126 select c1_0.co_code,c1_0.co_name from country c1_0
29-06-26 23:24:12.090 restartedMain      DEBUG c.c.o.OrmlLearnApplication      testGetAllCountries      38 countries=[Country [code=IN, name=India], Country [
code=US, name=United States of America]]
29-06-26 23:24:12.090 restartedMain      INFO c.c.o.OrmlLearnApplication      testGetAllCountries      39 End - Country Test
29-06-26 23:24:12.090 restartedMain      INFO c.c.o.OrmlLearnApplication      testAddEmployee      43 Start - Employee Add Test
29-06-26 23:24:12.090 restartedMain      DEBUG c.c.o.OrmlLearnApplication      testAddEmployee      48 Saving employee: Aditya Yadav
29-06-26 23:24:12.102 restartedMain      DEBUG org.hibernate.SQL      logStatement      126 insert into employee (em_name) values (?)
29-06-26 23:24:12.122 restartedMain      INFO c.c.o.OrmlLearnApplication      testAddEmployee      52 Employee saved successfully! Generated ID: 1
29-06-26 23:24:12.122 restartedMain      INFO c.c.o.OrmlLearnApplication      testAddEmployee      53 End - Employee Add Test
29-06-26 23:24:12.126 licationShutdownHook INFO rEntityManagerFactoryBean      destroy      693 Closing JPA EntityManagerFactory for persistence un
it 'default'
29-06-26 23:24:12.129 licationShutdownHook INFO c.z.h.HikariDataSource      close      349 HikariPool-1 - Shutdown initiated...
29-06-26 23:24:12.135 licationShutdownHook INFO c.z.h.HikariDataSource      close      351 HikariPool-1 - Shutdown completed.
```